An Undergrad-Friendly Ideal-Theoretic Calculation for a Two-Row Springer Cell Closure

Notes for the $N = 8$ example

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Abstract

This note walks through one concrete ideal-theoretic calculation for the intersection of the closure of one affine Springer Schubert cell with another. The example is the two-row Springer fiber of type $(4, 4)$, with source matching

$$M = \{(1, 8), (2, 7), (3, 4), (5, 6)\}$$

and target matching

$$F = \{(1, 2), (3, 4), (5, 6), (7, 8)\}.$$

The final answer is

$$\overline{C_M} \cap C_F = \{s = p\} \subset C_F.$$

The calculation is done twice. First, we use all off-cell Plucker coordinates that vanish on the target cell. Then we explain why, for the final target-cell ideal, it is enough in this example to keep only two off-cell coordinates X and Y.

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1 The point of the method

Suppose C_M and C_F are two affine cells in a Springer fiber. The problem is to compute

$$\overline{C_M} \cap C_F$$

as a subset of the affine target cell $C_F \cong \mathbb{A}^d$.

The dangerous mistake is to impose the equations of the target cell too early. Usually no finite point of the source cell lies in the target cell. The intersection appears only after taking a closure. The correct order is:

$$\boxed{\text{source cell} \longrightarrow \text{target chart} \longrightarrow \text{take closure} \longrightarrow \text{impose target-cell equations.}}$$

The algebraic version is:

Step 1. Write the source cell in coordinates.

Step 2. Choose the target flag chart.

Step 3. Express the target-chart Plucker ratios on the source cell.

Step 4. Clear denominators and form a graph ideal.

Step 5. Saturate by the product of the denominators.

Step 6. Eliminate the source variables.

Step 7. Finally impose the equations cutting out the target cell inside the target chart.

2 The source and target matrices

We use the source matching

$$M = \{(1,8),(2,7),(3,4),(5,6)\}$$

and the target matching

$$F = \{(1,2),(3,4),(5,6),(7,8)\}.$$

The source cell is parametrized by

$$G(a,b,c,d) = \begin{pmatrix} a & b & c & 1 & 0 & 0 & 0 & 0 \\ 0 & a & b & 0 & d & 1 & 0 & 0 \\ 0 & 0 & a & 0 & b & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & a & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \end{pmatrix}.$$

The fully unnested target cell is parametrized by

$$H(p,q,r,s) = \begin{pmatrix} p & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & q & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & r & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & s & 1 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

A point of the flag variety is represented by the sequence of column spans

$$V_i = \text{span(first } i \text{ columns)}.$$

For each i, its Plucker coordinates are the $i \times i$ minors of the first i columns.

3 The target chart

The pivot rows of the target matrix $H(p,q,r,s)$ are

$$5, 1, 6, 2, 7, 3, 8, 4.$$

Therefore the pivot sets for the first i columns are

$$\begin{aligned} R_1 &= \{5\}, \\ R_2 &= \{1,5\}, \\ R_3 &= \{1,5,6\}, \\ R_4 &= \{1,2,5,6\}, \\ R_5 &= \{1,2,5,6,7\}, \\ R_6 &= \{1,2,3,5,6,7\}, \\ R_7 &= \{1,2,3,5,6,7,8\}. \end{aligned}$$

The target flag chart is

$$U_F = \bigcap_{i=1}^{7} D\left(P^{(i)}_{R_i}\right).$$

Here $P^{(i)}_I$ denotes the $i \times i$ minor using row set I and the first i columns.

When these pivot minors are evaluated on the source matrix $G(a,b,c,d)$, one gets, up to signs,

$$1, \quad b, \quad c, \quad b, \quad d, \quad b, \quad 1.$$

Thus the relevant denominator product is

$$\Delta = bcd.$$

The full product has a higher power of b, but saturating by b^3cd is the same as saturating by bcd.

4 Target-cell coordinates

We choose four chart ratios that become the target variables p, q, r, s on the target cell:

$$P = \frac{P_1^{(1)}}{P_5^{(1)}},$$

$$Q = -\frac{P_{125}^{(3)}}{P_{156}^{(3)}},$$

$$R = \frac{P_{12356}^{(5)}}{P_{12567}^{(5)}},$$

$$S = -\frac{P_{1234567}^{(7)}}{P_{1235678}^{(7)}}.$$

On the target matrix $H(p, q, r, s)$, these are exactly

$$P = p, \qquad Q = q, \qquad R = r, \qquad S = s.$$

On the source matrix $G(a, b, c, d)$, they are

$$\boxed{P = a, \qquad Q = a - \frac{b^2}{c}, \qquad R = a - \frac{b^2}{d}, \qquad S = a.}$$

These are the coupled rational functions that make the example interesting.

5 All off-cell coordinates

Inside the target chart U_F, the target cell C_F is obtained by requiring many other chart coordinates to vanish. These are the off-cell coordinates.

For each rank i, define

$$E_I^{(i)} = \frac{P_I^{(i)}}{P_{R_i}^{(i)}}.$$

The following table lists every such coordinate that vanishes identically on $H(p, q, r, s)$ but is nonzero on the source matrix $G(a, b, c, d)$.

Rank	Off-cell coordinates evaluated on $G(a, b, c, d)$
2	$E_{12}^{(2)} = -\frac{a^2}{b}, \quad E_{16}^{(2)} = -\frac{a}{b}, \quad E_{25}^{(2)} = \frac{a}{b}, \quad E_{56}^{(2)} = -\frac{1}{b}.$

Rank	Off-cell coordinates evaluated on $G(a,b,c,d)$
3	$$\begin{aligned} E^{(3)}_{123} &= \frac{a^3}{c}, & E^{(3)}_{126} &= -\frac{ab}{c}, & E^{(3)}_{127} &= \frac{a^2}{c}, \\ E^{(3)}_{135} &= \frac{ab}{c}, & E^{(3)}_{136} &= -\frac{a^2}{c}, & E^{(3)}_{157} &= -\frac{b}{c}, \\ E^{(3)}_{167} &= \frac{a}{c}, & E^{(3)}_{235} &= \frac{a^2}{c}, & E^{(3)}_{256} &= \frac{b}{c}, \\ E^{(3)}_{257} &= -\frac{a}{c}, & E^{(3)}_{356} &= \frac{a}{c}, & E^{(3)}_{567} &= \frac{1}{c}. \end{aligned}$$
4	$$E^{(4)}_{1235} = \frac{a^2}{b}, \quad E^{(4)}_{1257} = -\frac{a}{b}, \quad E^{(4)}_{1356} = \frac{a}{b}, \quad E^{(4)}_{1567} = \frac{1}{b}.$$
5	$$\begin{aligned} E^{(5)}_{12345} &= \frac{a^3}{d}, & E^{(5)}_{12357} &= \frac{ab}{d}, & E^{(5)}_{12358} &= -\frac{a^2}{d}, \\ E^{(5)}_{12456} &= -\frac{ab}{d}, & E^{(5)}_{12457} &= \frac{a^2}{d}, & E^{(5)}_{12568} &= -\frac{b}{d}, \\ E^{(5)}_{12578} &= \frac{a}{d}, & E^{(5)}_{13456} &= -\frac{a^2}{d}, & E^{(5)}_{13567} &= \frac{b}{d}, \\ E^{(5)}_{13568} &= -\frac{a}{d}, & E^{(5)}_{14567} &= \frac{a}{d}, & E^{(5)}_{15678} &= -\frac{1}{d}. \end{aligned}$$
6	$$E^{(6)}_{123456} = -\frac{a^2}{b}, \quad E^{(6)}_{123568} = -\frac{a}{b}, \quad E^{(6)}_{124567} = \frac{a}{b}, \quad E^{(6)}_{125678} = -\frac{1}{b}.$$

So the full target-cell condition inside this chart is:

$$\boxed{\text{all coordinates in the table are zero.}}$$

6 A smaller set of normal coordinates

Although the table looks large, all of these coordinates are controlled by three simple normal variables.

Set

$$\tau = \frac{1}{b}, \qquad X = \frac{b}{c}, \qquad Y = \frac{b}{d}.$$

Here X and Y are themselves Plucker ratios:

$$X = E^{(3)}_{256}, \qquad Y = E^{(5)}_{13567}.$$

Also, up to sign, τ is the rank-two off-cell coordinate $E^{(2)}_{56}$:

$$E^{(2)}_{56} = -\tau.$$

Every off-cell coordinate in the table is a polynomial in P, τ, X, Y. For instance,

$$\frac{a^2}{b} = P^2\tau, \qquad \frac{a}{c} = PX\tau, \qquad \frac{ab}{c} = PX, \qquad \frac{b}{d} = Y, \qquad \frac{1}{d} = Y\tau.$$

Consequently, on the source-chart closure, imposing all off-cell coordinates to vanish is equivalent to imposing

$$\boxed{\tau = X = Y = 0.}$$

This is the full formulation. In the simplified formulation, we keep only X and Y because they are enough to recover the final ideal in the target variables P, Q, R, S. The missing variable τ imposes no additional equation among P, Q, R, S after elimination.

7 The graph ideal and saturation

The rational formulas are

$$P = a, \qquad Q = a - \frac{b^2}{c}, \qquad R = a - \frac{b^2}{d}, \qquad S = a,$$

$$X = \frac{b}{c}, \qquad Y = \frac{b}{d}.$$

Clearing denominators gives the graph ideal

$$J = \Big\langle P - a, S - a, c(Q-a) + b^2, d(R-a) + b^2, cX - b, dY - b \Big\rangle$$

in the ring

$$\mathbb{C}[a, b, c, d, P, Q, R, S, X, Y].$$

This graph is only meaningful on the open set

$$D(bcd) = \{bcd \neq 0\}.$$

Therefore the correct ideal is not J, but the saturation

$$J^{\text{sat}} = J : (bcd)^\infty.$$

What saturation means

For an ideal $I \subset R$ and a polynomial $f \in R$, the saturation is

$$I : f^\infty = \{g \in R : f^k g \in I \text{ for some } k \gg 0\}.$$

Geometrically, over $\mathbb{C}$,

$$V(I : f^\infty) = \overline{V(I) \cap D(f)}.$$

In words:

$$\boxed{\text{saturation removes fake components introduced by clearing denominators.}}$$

In this example, without saturation, the equations allow the fake component

$$b = c = d = 0, \qquad P = S = a,$$

with Q, R, X, Y arbitrary. This fake component is not part of the original rational graph, because the rational functions were not defined when $bcd = 0$.

8 The saturated elimination calculation

Eliminating the source variables a, b, c, d from J^{sat} gives

$$I_{\text{chart}} = \langle S - P,\ Y(P-Q) - X(P-R) \rangle \subset \mathbb{C}[P, Q, R, S, X, Y].$$

The relation is easy to understand. On the source open set,

$$P - Q = \frac{b^2}{c} = bX,$$

and

$$P - R = \frac{b^2}{d} = bY.$$

Therefore

$$Y(P-Q) = YbX = XbY = X(P-R).$$

This gives the determinantal relation

$$Y(P-Q) - X(P-R) = 0.$$

Thus the closure of the source cell in the target chart is cut out by

$$\boxed{S = P, \qquad Y(P-Q) = X(P-R).}$$

9 Full formulation: impose all vanishing coordinates

The full target cell condition says every off-cell coordinate in the table is zero. On the source-chart closure, this is equivalent to

$$\tau = X = Y = 0.$$

The full local chart ideal can be written as

$$I_{\text{full}} = \langle S - P,\ X - \tau(P-Q),\ Y - \tau(P-R) \rangle \subset \mathbb{C}[P, Q, R, S, \tau, X, Y].$$

Indeed,

$$P - Q = \frac{b^2}{c} = bX = \frac{X}{\tau},$$

so

$$X = \tau(P-Q),$$

and similarly

$$Y = \tau(P-R).$$

Now impose the full target-cell condition:

$$\tau = X = Y = 0.$$

Eliminating τ, X, Y leaves

$$\boxed{S - P = 0.}$$

Therefore

$$\boxed{\overline{C_M} \cap C_F = \{s = p\} \subset C_F.}$$

10 Simplified formulation: only use X and Y

The simplified chart ideal is

$$I_{\text{chart}} = \langle S - P,\ Y(P - Q) - X(P - R)\rangle\,.$$

Now impose only

$$X = Y = 0.$$

The determinant relation becomes automatic, and the only equation left among the target variables is

$$S - P = 0.$$

So the simplified computation gives the same target-cell ideal:

$$\boxed{\overline{C_M} \cap C_F = \{s = p\}.}$$

Why this is allowed here

The full target-cell condition really requires all off-cell coordinates to vanish, equivalently $\tau = X = Y = 0$. But the extra equation $\tau = 0$ only lives in the forgotten normal direction. After we eliminate the normal variables and keep only the target variables P, Q, R, S, it imposes no additional equation. Thus, for the final ideal in $\mathbb{C}[P, Q, R, S]$, it is enough in this example to impose $X = Y = 0$.

This is a useful shortcut for hand calculations, but for computer calculations one should usually include all off-cell coordinates first, then simplify afterward.

11 Why saturation matters

If we eliminate from the unsaturated graph ideal J, we only get

$$J \cap \mathbb{C}[P, Q, R, S, X, Y] = \langle S - P\rangle\,.$$

This is too large as a chart closure because it misses the relation

$$Y(P - Q) - X(P - R) = 0.$$

After saturation, we get the correct chart closure:

$$(J : (bcd)^\infty) \cap \mathbb{C}[P, Q, R, S, X, Y] = \langle S - P,\ Y(P - Q) - X(P - R)\rangle\,.$$

So saturation is not cosmetic. It removes denominator-clearing artifacts.

12 Macaulay2 code

Here is a minimal Macaulay2 script for the simplified calculation.

```
R = QQ[a,b,c,d,P,Q,Rr,S,X,Y, MonomialOrder => Lex];

J = ideal(
    P-a,
    S-a,
```

```
    c*(Q-a)+b^2,
    d*(Rr-a)+b^2,
    c*X-b,
    d*Y-b
);

Delta = b*c*d;

-- Incorrect: no saturation
Ibad = eliminate({a,b,c,d}, J)

-- Correct: saturated graph closure
Jsat = saturate(J, ideal(Delta));
Ichart = eliminate({a,b,c,d}, Jsat)

-- Intersect target cell using X=Y=0
Itarget = eliminate({X,Y}, Ichart + ideal(X,Y))
```

Expected output:

```
Ibad = ideal(S-P)

Ichart = ideal(S-P, Y*(P-Q)-X*(P-R))

Itarget = ideal(S-P)
```

If you want to avoid using the variable name R because it conflicts with the ring name, use Rr as above.

13 Undergrad-friendly recipe

Here is the recipe I would give to a student.

Input

You need:

- a source cell matrix $G_M(\mathbf{t})$,
- a target cell matrix $G_N(\mathbf{s})$.

The source variables are $\mathbf{t}$ and the target variables are $\mathbf{s}$.

Step 1: Find the target pivots

Look at the target matrix. For each column, identify its pivot row. Then form the prefix pivot sets

$$R_i = \{\text{pivot rows of the first } i \text{ columns}\}.$$

Step 2: Choose target chart coordinates

For each rank i, divide by the pivot minor $P^{(i)}_{R_i}$. A chart coordinate looks like

$$\frac{P^{(i)}_{I}}{P^{(i)}_{R_i}}.$$

Choose:

- coordinates that become the target variables on $G_N(\mathbf{s})$,
- coordinates that vanish on $G_N(\mathbf{s})$; these are off-cell coordinates.

Step 3: Evaluate on the source

Substitute the source matrix $G_M(\mathbf{t})$ into those ratios. You will get rational functions.

Example:

$$Q = a - \frac{b^2}{c}, \qquad X = \frac{b}{c}.$$

Step 4: Clear denominators

If

$$Z = \frac{A(\mathbf{t})}{D(\mathbf{t})},$$

write instead the polynomial equation

$$D(\mathbf{t})Z - A(\mathbf{t}) = 0.$$

The ideal generated by these equations is the graph ideal J.

Step 5: Saturate

Let Δ be the product of the denominators. Compute

$$J^{\text{sat}} = J : \Delta^{\infty}.$$

This removes fake components lying where denominators vanish.

Step 6: Eliminate source variables

Compute

$$I_{\text{chart}} = J^{\text{sat}} \cap \mathbb{C}[\text{target chart variables}].$$

This is the closure of the source cell inside the target chart.

Step 7: Intersect the target cell

Now impose all off-cell coordinates equal to zero. Finally eliminate any off-cell variables you do not want to keep. The resulting ideal in the target variables $\mathbf{s}$ defines

$$\overline{C_M} \cap C_N \subset C_N.$$

Step 8: Interpret the answer

Common outputs are:

Ideal	Meaning
(1)	empty intersection
(0)	the whole target cell is contained in the closure
linear equations	an affine-linear cut locus
nonlinear equations	interesting geometry; check carefully
non-radical ideal	possible scheme structure

14 Takeaway

The example illustrates three lessons.

1. The closure must be computed in a target chart before imposing the target-cell equations.
2. Saturation is essential: it removes fake components created by clearing denominators.
3. In hand calculations, a few carefully chosen normal coordinates may be enough. In this example, $X = b/c$ and $Y = b/d$ recover the final target-cell ideal, even though the full target-cell condition requires all off-cell coordinates to vanish.